

Fundamentals of Machine learning for and with engineering applications

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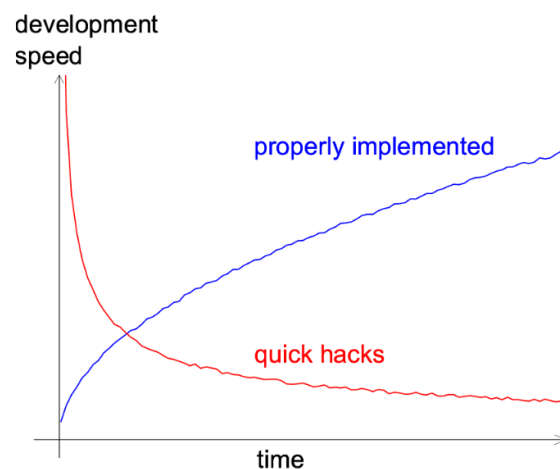
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Flexible



Developing approaches

Different code editors are available to interpret python language.

- jupyter notebooks are mostly dedicated to learning (Markdown)
- ipython is for interactive coding (similar to R, Matlab, etc)
- python packages (.py) developing suites (debug possibilities and git integration)

Introducing code standards

When developing code, there are **guidelines** and best practices aimed at improving the **quality, readability, and maintainability** of a code.

There are different levels of coding quality, mostly depending on the code intended usage (and developer skills).

- Private codes can be whatever (Cpt. Obvious)
- Public packages shall use a 'Golden code standards' such to be used and eventually supported by communities.

Community level standards

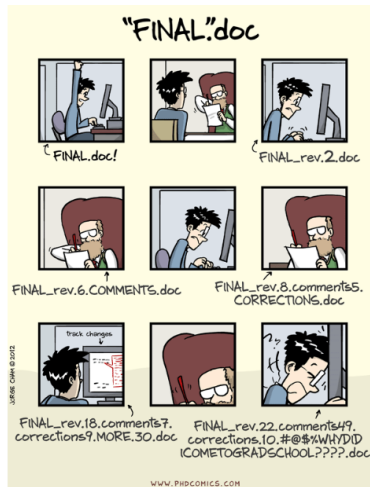
Principle of 'clean coding':

- 1 Readability and Clarity: A good code shall be possible to read as when reading a book
- 2 Structure and object oriented: A code shall be composed by objects, each of them connected in the less redundant way possible.
- 3 Consistency and Style: Variable naming, function naming and classes naming has to be consistent.
- 4 Documentation: Each file, each function and each class shall contain the relative description of its aim and its usage
- 5 Maintainability: Code dependencies have to be stated and consistently defined and updated, such that a suitable environment can be developed at any point in time.

Community level standard

- 1 Testing: Unit testing shall cover the majority of the code
- 2 Error Handling: Each error shall be captured and properly identified.
- 3 Examples and benchmarks: Users shall be able to execute minimal examples of the code for computational checks.
- 4 Performance Optimization: Libraries shall be able to use the available computational power in the machine (e.g. GPU-CUDA)

Version control



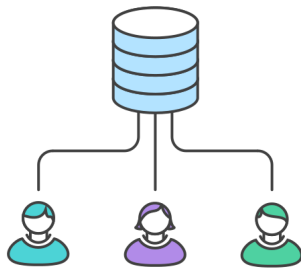
Git

Git is a distributed version control system that tracks changes in any set of computer files, usually used for coordinating work among programmers who are collaboratively developing source code during software development. Its goals include speed, data integrity, and support for distributed, non-linear workflows (thousands of parallel branches running on different computers). [Wiki]

Let's try to be more accessible.

Git is a computer program/tool to save and download files on a hosting server (e.g. GitHub and GitLab).

Centralized workflow



A distributed version control system

GIT

- Git facilitates users to track the various versions of files. It is not a necessary tool, but it can be very very helpful. Generally, the time spent to learn its syntax is well paid off

(do you remember to save some file like *manuscript_draft_v4.02_final_definitive_forreal_lastcomments_edited* Exactly! Imagine to do that for a repository of files...)

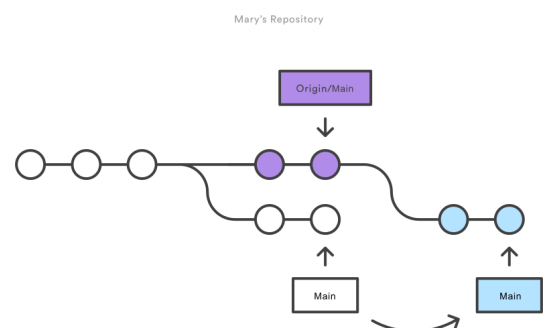
- It permits to save and share the intermediate stages of a work in progress (which software is complete and always up to date?) in an accessible, consistent and structured way, allowing an effective version tracking. It allows retrieval of previous working versions, limiting the risk to overwrite useful files.

What is git actually for

The tool is particularly useful for programmers working in teams or in projects whose outcomes can be used by others.

- Git helps to co-develop a code, test its functions and the compatibility of the various code sections.
- A long list of further possibilities became possible by git.
- Different software integration on development platforms, based on git, will help you to develop and co-develop your code.
- The platform GitLab and GitHub have a large set of functionalities to further support code documentation and public releases.
- Files can be disclosed to the public, becoming a great integration of your CV, showing what you are able to do in an open and accessible way.

How does it work -in short-



Why should I care?

As the open libraries are exploding in numbers, you might need some criteria to assert the reliability of a project.

Unit test driven development!

That is taking full advantage of python object oriented structure.

Community

Good project are not only used by communities, but also **supported**

Git allows the development of projects without a clear lead.

Community engagement is generally a desirable target to help develop to directly integrate feedbacks by users (and fix bugs).

It can help your CV! maybe...

**"you can find the
projects I worked on
on my GitHub"
My GitHub:**

